Pooja Kiran Bharadwaj

Security Engineering | Detection & Response | AI Security | Cloud Security

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Greater Phoenix Area, AZ | F-1 OPT EAD | Open to relocation

Security engineer focused on detection engineering, security telemetry, threat analysis, and automated response for cloud and AI infrastructure. Build security systems that normalize runtime events, identify attacker behavior, correlate multi-stage attack activity, enforce security decisions, and produce auditable evidence. Hands-on experience with Elastic SIEM, Kubernetes, AWS IAM, security automation, MITRE ATT&CK, adversarial testing, and AI security. Built and maintain open-source security systems spanning agent runtime security, cloud IAM attack-path analysis, model supply-chain security, and adversarial detection.

Experience

Independent AI Security Researcher & Engineer — Aug. 2024 – Present
Self-Directed Security Research (concurrent with M.S. program) — *Tempe, AZ*

- Design and maintain 16 open-source security repositories spanning detection engineering, security enforcement, threat analysis, cloud IAM, AI agent security, model supply-chain security, and adversarial testing.
- Engineer detection pipelines that normalize security events, evaluate attacker behavior, correlate multi-stage activity, classify findings by severity, map findings to MITRE ATT&CK and MITRE ATLAS, and export SARIF for CI security workflows.
- Build automated security controls that make allow, block, redact, quarantine, and deny decisions with circuit breakers, rate limiting, tamper-evident audit logging, and explicit failure-mode handling.
- Validate security controls through automated unit, integration, failure-mode, and adversarial tests, including 569 tests for the MCP security gateway and 166 tests for the AWS IAM analyzer.
- Integrate CodeQL, Bandit, Trivy, Grype, Syft SBOM, pip-audit, and multi-version CI to continuously validate security and software supply-chain controls.

Business Compliance Lead & Market/Cost Analyst — Aug. 2025 – Dec. 2025
ASU Technology Innovation Lab (Honeywell Aerospace partnership) — *Tempe, AZ*

- Evaluated cybersecurity product requirements against aerospace compliance frameworks and translated security risks into system-level architecture constraints.
- Conducted competitive and market analysis to evaluate security requirements, product capabilities, and implementation tradeoffs.

Graduate Teaching Assistant, IT Grader — Jan. 2025 – Oct. 2025
Ira A. Fulton Schools of Engineering, Arizona State University — *Mesa, AZ*

- Assessed cloud security architectures, network defense configurations, and IAM implementations for 100+ students per semester.
- Provided structured technical feedback on security architecture, implementation quality, threat exposure, and defensive controls.

Security Engineering Projects

MCP Agent Security Gateway *(17 stars)* | *Python, FastAPI, Docker, Kubernetes* — Jan. 2025 – Aug. 2026

- Inline security proxy that intercepts MCP agent-to-tool calls via JSON-RPC stdio, evaluates each request through a 5-layer detection pipeline (server trust, tool-call policy, process-spawn analysis, semantic intent, network egress), and returns allow/block/redact/quarantine decisions before requests reach downstream servers.
- Detection engine: 50+ prompt-injection rules with Unicode normalization, zero-width character stripping, homoglyph detection, Base64/ROT13 decode; PII detection across 9 data types; exfiltration-pattern matching for hidden recipients, oversized payloads, and raw-IP destinations.
- Operational controls: SHA-256 hash-chained audit logging, write-ahead log for security decisions, circuit breakers (fail-closed on breaker open), rate limiting, shadow mode for safe detection rollout, W3C traceparent propagation, Prometheus metrics. Kubernetes deployment manifests with HPA, liveness/readiness probes, and security contexts.
- Built a detection engineering lab: exports security decisions in Elastic Common Schema (ECS), ships to Elasticsearch/Kibana via Filebeat, and runs an in-memory correlation engine with 6 multi-event rules that detect attack chains (recon to exfil, injection to privilege escalation) invisible in single events. Includes 9 Elastic Security detection rules and 6 Atomic Red Team style attack simulations, all mapped to MITRE ATT&CK. Validated locally against a docker-compose ELK stack.
- 569 tests, 75% coverage across Python 3.10-3.12. CI includes CodeQL, Bandit, Trivy, Grype, Syft SBOM, pip-audit, Docker image build. Tested on both Linux (CI) and Windows.

AWS Agent Identity Guard | *Python, AWS IAM, Terraform, SARIF* — Mar. 2025 – Aug. 2026

- Static analyzer that detects dangerous IAM permission patterns in AI agent roles. 25 rules covering: wildcard service grants, iam:PassRole without conditions, privilege-management actions, audit-trail tampering (CloudTrail, GuardDuty, Config), credential-harvest chains, lateral-movement combinations, cross-account trust without ExternalId.
- Multi-step attack-path detection: identifies when a single identity combines credential harvesting, metadata reachability, and lateral-movement permissions into an exploitable chain. This catches privilege escalation that single-rule scanners miss.
- Outputs Text, JSON, and SARIF. Exit code 1 on critical/high findings serves as a CI merge gate. Integrates with GitHub Code Scanning for inline PR annotations. Optional live-account scanning via boto3 (read-only IAM APIs). 166 automated tests including failure-mode coverage.

HF Model Provenance Scanner | *Python, Hugging Face API, SARIF* — Sep. 2024 – Aug. 2026

- Supply-chain scanner for Hugging Face model repositories. Analyzes pickle opcodes (Protocol 0-5), SafeTensors headers, GGUF structures, ONNX, and Keras files without downloading full weights. Detects malicious deserialization payloads, typosquatting, obfuscated code, and provenance gaps.
- Implements temporal baseline diffing for detecting rug-pull modifications, Levenshtein-based typosquat detection, multi-layer obfuscation decode (base64, chr(), string concatenation), and CycloneDX 1.6 SBOM generation. All findings mapped to MITRE ATT&CK v19 techniques.

LLM Red-Team Framework | *Python, scikit-learn, FastAPI, SARIF* — Feb. 2025 – Jul. 2026

- Offline adversarial evaluation harness for prompt-injection detectors. Generates attack corpora across 6 categories mapped to OWASP LLM Top 10, trains a baseline TF-IDF classifier, and measures performance using grouped template splits that prevent data leakage. F1 = 0.97 on held-out templates; honestly reports degradation on novel phrasings.
- FastAPI service exposes /scan endpoint with rate limiting and API-key auth. SARIF output integrates with GitHub Code Scanning. 84 tests, 94% coverage.

Education

Arizona State University — Tempe, AZ
M.S. Information Technology — *Aug. 2024 – May 2026*

- Relevant coursework: Machine Learning Security, Cloud Security Architecture, Information Assurance, Network Defense

M. S. Ramaiah University of Applied Sciences — Bengaluru, India
B.Tech Computer Science & Engineering — *Aug. 2019 – Aug. 2023*

Technical Skills

Detection & Response: Detection engineering, SIEM integration, Elastic/ELK, ECS, event correlation, attack-chain detection, security telemetry, threat analysis, attack simulation, threat modeling, audit logging, policy enforcement, security automation, SARIF
Cloud & Infrastructure: AWS IAM, Kubernetes, Docker, Terraform, GitHub Actions, Elasticsearch, Kibana, Filebeat, Prometheus, Grafana
AI Security: AI agent runtime security, prompt-injection detection, model supply-chain security, LLM security evaluation, adversarial ML, data poisoning detection, adversarial robustness
Threat Intelligence & Frameworks: MITRE ATT&CK v19, MITRE ATLAS, OWASP Top 10 for LLMs, NIST AI RMF, CycloneDX SBOM
Languages & Frameworks: Python, SQL, FastAPI, PyTorch, scikit-learn, NumPy
Security Tooling: CodeQL, Bandit, Trivy, Grype, pip-audit, Syft, Dependabot

Publication & Recognition

IEEE INDICON 2023: “A Personalized E-Learning System Using Reinforcement Learning Through Satellite” (IEEE Xplore: 10440852)
AWS Academy Graduate: Cloud Security Foundations
KSCST Research Grant: Funded undergraduate research project